Merge Linked Lists

AlgoExpert · Hard · Linked Lists

The problem

- Merge two sorted lists in place; return the head of the merged list.

Intuition

- Walk both sorted lists, splicing the smaller node in front each time, without new nodes.

Approach

- 1. Keep pointers into both lists and a previous pointer
- 2. Splice the smaller node
- 3. Attach any leftover tail
- 4. Return the smaller head

Java solution

```java
public class MergeLinkedLists {
    public static LinkedListNode mergeLinkedLists(LinkedListNode a, LinkedListNode b) {
        LinkedListNode dummy = new LinkedListNode(0), t = dummy;
        while (a != null && b != null) {
            if (a.value <= b.value) { t.next = a; a = a.next; } else { t.next = b; b = b.next; }
            t = t.next;
        }
        t.next = a != null ? a : b;
        return dummy.next;
    }
}
```

Complexity

- Time: O(n + m)
- Space: O(1)

Recap

- Walk both sorted lists, splicing the smaller node in front each time, without new nodes.
- Complexity: O(n + m)